Name.....

Signature.....**Candidates Personal Number**.....

545/1

CHEMISTRY.

Paper 1.

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Uganda Certificate of Education

Internal mock set 9

Paper 1

2 hours

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part.. Answer **four** items in all.

Answers to section **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored.*

SECTION A

Answer **all** *items in this section in the spaces provided*

Item 1

A local investor is planning to establish a beverage processing factory on a small scale. For high quality production, a friend has advised him to use specific ingredients during the production process. These include **citric acid**, a common preservative extracted from lemon ,**vanilla** obtained from cured unripe fruit of vanilla orchid and **honey**. Seeking further guidance on these substances, he has consulted you for expert guidance.

Task

As a learner of chemistry,

(a) State the category of the substances he was advised to use.

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(b) How does the substances ensure the quality of the product?

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(c) explain the dangers associated with the use of the substances.

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(d) Compare the substances he was advised to use with **other type** of substances he would alternatively use.

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Item 2.

As part of the country's effort to transition to a low carbon future, the officials from ministry of energy have initiated a comprehensive education campaign to the public about the nature and use of two substances, **biogas** and **natural gas** as better alternatives in the energy landscape. This initiative will emphasize the need for the establishment of actionable strategies for the sustainable use of these substances in our daily lives. You have been invited to one of the country's most listened to radio, to educate the public.

Task: As a Chemistry student,

(a) explain the category of the substances.

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(b) suggest the properties of **any one** substance that make it useful in life.

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(c) advise on the environmental impact associated with use of **any one** substance.

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SECTION B

Part I (*Attempt one item from this section)*

Item 3 Aluminium is a lightweight versatile metal used for making bodies of cars, kitchen pans and for electrical wiring among others .To increase on quantities of the metal on market needed , the government has cleared an investor to set up an aluminium extraction plant in one district. However ,the locals are concerned about its impact and are seeking guidance on how they will benefit from it's operations. The officials have organized a workshop to educate the locals. You have been invited to attend.

Task.

As a chemistry student, make a write up you will use during the presentation.

Item 4

Due to large quantities of ethanol needed on market for medical and pharmaceutical applications, production of alcoholic beverages and personal care products, Masaka city officials have cleared an investor to set up an ethanol manufacturing plant. However, the locals are concerned about its impact to the environment and how they will socially benefit from the plant. The officials have organized a workshop at the city hall. You have been invited to address the locals.

Task. Prepare a write up message you will use.

Part II

(*Attempt* **one** *item* from *part)*

Item 5

The industrial sector's contribution to the national economy and job creation is undeniable. However, this economic progress has often come at the expense of environment as observed by it's signifant impact on water and and air quality. This has raised concern among many environmentalists. They have organized a radio talk show involving all stake holders and owners of different industries, both small and large scale. You have been invited to attend.

Task. Prepare a write up message you will deliver.

Item 6

Uganda's recent discovery of large crude oil reserves within the Albertine region in Hoima presents a significant opportunity to meet it's energy needs. However, this development has sparked considerable environmental concern among the locals. This stems from the potential impact of the extraction and refinery processes on waters of lake Albert and nearby community water sources . Officials from ministry of energy have organized workshops to address the locals . You have been invited to address the locals .

Task.

Write a write up presentation you will use during the presentation.

End.